

JAKE ALLYN

Physicist | Simulation & Software Engineer

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SUMMARY

I'm a physicist first — I did research in galactic dynamics as an undergrad and spent two years running blast and debris experiments before ever writing production code. That background is still what drives my work today: whether it's orbital mechanics, sensor physics, or vehicle dynamics, I'm usually starting from the physical model and building the software around it, not the other way around. That approach has carried me from peer-reviewed structural research, to four years shipping real-time vehicle physics on NBA 2K, to my current work modeling sensor and environmental physics for the Collaborative Combat Aircraft program. Currently pursuing DoD security clearance adjudication.

CORE COMPETENCIES

Computational Physics • Orbital & Rigid-Body Dynamics • Sensor & Environmental Physics Modeling • N-Body Simulation • Blast & Debris Mechanics • Monte Carlo / Statistical Modeling • Modeling & Simulation (M&S) • Real-Time Simulation • C++ • Python • MATLAB • Unreal Engine • Unity • Godot • JSBSim • QGIS • Security Clearance (In Process)

EXPERIENCE

Staff Software Developer — Physics & Simulation

May 2026 – Present

General Atomics Aeronautical Systems | *Poway, CA (Hybrid)*

- Physics specialist on the Autonomy and Simulation team for the Collaborative Combat Aircraft (CCA) program, FQ-42A, developing the physical models underlying the Simulation Platform and Vehicle Isolator.
- Modeled the underlying physics of infrared detection and atmospheric/environmental effects to build an Infrared Search and Track (IRST) sensor simulation and weather/environment simulation system.
- Pursuing DoD security clearance adjudication.

Lead Physics and Simulation Engineer

Nov 2025 – May 2026

e-bliss LLC | *Rancho Palos Verdes, CA (Remote)*

- Designed and implemented a custom orbital mechanics simulation engine in C++, applying orbital dynamics and propagation theory to satellite mission modeling and constellation-level simulation.
- Modeled line-of-sight geometry constraints for satellite-to-satellite and satellite-to-ground communication links.
- Built a Godot-based GUI for scenario visualization and configuration, and developed tools for evaluating mission feasibility and communication coverage.
- Acted as project manager for a government proposal effort, identifying and evaluating SBIR solicitations and BAAs aligned with company technical capabilities.

Physics Software Engineer

Sep 2022 – Mar 2026

Visual Concepts | *Novato, CA (Remote)*

- Designed and implemented a modular vehicle physics system grounded in rigid-body dynamics, suspension modeling, and tire/traction physics, supporting tanks, cars, aircraft, and hovercraft with custom handling profiles across multiple NBA 2K titles.
- Led racetrack physics and gameplay development, including AI behavior and environmental interactions; owned vehicle content delivery across NBA 2K24, 2K25, and long-term vehicle planning for 2K26.
- Refactored the full input system for reliability and debuggability, and built cross-platform input visualization and remapping tooling.
- Built a dynamic prop-loading system to streamline content creation and enable flexible venue customization.
- Profiled and optimized the inverse kinematics system for NPCs, reducing frame impact and enabling scalable LODs.

Computational Physicist

Dec 2020 – Sep 2022

ACTA-SH | *Torrance, CA (On-site)*

- Led large-scale experiments analyzing structural failure and debris propagation; designed and executed blast and ballistic tests on CMU and RC walls to measure debris impact on internal equipment.
- Developed reusable impulse sleds to capture secondary debris data that became standard tooling in related testing domains.
- Partnered with the FAA and USAF to build simulation models of aircraft failure scenarios and ground-level debris damage; built custom visualization tools used in formal safety assessments and stakeholder presentations.

- Maintained and modernized legacy simulation tools in Python, C++, and Fortran; built Monte Carlo pipelines for large-scale risk and dispersion analysis and supported peer-reviewed publications through figure generation and reproducible experiment pipelines.

Undergraduate Research Assistant, Physics

Sep 2016 – Jun 2019

UC Irvine | Irvine, CA

- Conducted research under Prof. Manoj Kaplinghat modeling gravitational interactions between the Milky Way, Andromeda, and satellite galaxies, using Python and C++ N-body and point-mass simulations (Rebound, GADGET-2, AMUSE).
- Analyzed simulation output to explain observed orbital patterns between galaxies, forming the basis of coursework and later publication work.

Earlier Experience

Full Stack & Unity/Mobile Development

- Godzilab Inc — Full Stack Developer (2018–2020): Built and maintained Python/JavaScript backend and support tooling for the mobile game "Crush Them All."
- The Ulysses Group / HFC Media — Lead Android/iOS Developer, Unity (2019–2020): Led cross-platform Unity development, including AR features for a holographic media application.

Personal Projects

- Building an independent Unity project modeling military vehicle combat systems, alongside procedural world generation and procedural first-person animation work.

EDUCATION

B.S., Physics — University of California, Irvine (2015 – 2019)

PUBLICATIONS

Shock and Vibration Symposium (2021–2022)

- Secondary Debris Effects on Building Infrastructure Derived from Experiments
- A Green's Function Methodology for Higher-Fidelity Surrogate Models
- Secondary Debris Impulse Measurements from Failing CMU and RC Walls